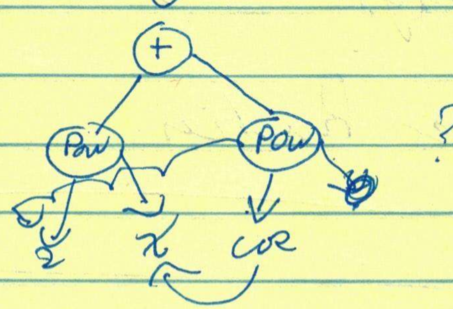
72D
rand notes

6/14

Xreplace: $x^2 + \cos(x) \xrightarrow[\sin x]{\cos x} x^2 + \sin(x)$

$x^2 + \cos^2(x) \xrightarrow[\sin x]{\cos x} x^2 + \sin^2(x)$? I guess yes since



$x^2 + \sin(\cos(x)) \xrightarrow[\sin x]{\cos x} x^2 + \sin(\sin(x))$

$x^2 + \cos(x) \xrightarrow{x \to \frac{\pi}{4}} \frac{\pi^2}{16} + \frac{\sqrt{2}}{2}$

subs: $x^2 + \cos(x) \xrightarrow[\sin x]{\cos x} x^2 + \sin(x)$

$4 \cdot x^4 + 3x^3 + 2x^2 + x \xrightarrow{x^2 \to y} 3x^3 + x + 4y^2 + 2y$

(since $x^2 \to y$ but $x \to \sqrt{y}$ or $-\sqrt{y}$)

usually not

Then subs & xreplace look almost the same except subs distinguishes between free & bound symbols

~~expr = Sym Integral(x·y,~~

$\int_0^1 dx\, x \cdot y \xrightarrow[y \to t]{x \to z} \int_0^1 dz\, z \cdot y$ (scribbled out)

$\xrightarrow{\text{xreplace}} \int_0^1 dz\, z \cdot t$

$\xrightarrow{\text{subs}} \int_0^1 dx\, x \cdot t$

$\Rightarrow$

a = Wild('a')

~~exp~~ $\sin(x) + \sin(y) \xrightarrow{\sin \to \cos} \cos(x) + \cos(y)$

replace also takes function

can combine

$$\boxed{4x^4 + 3x^3 + 2x^2 + x + \cos(x) + \frac{\cos^2(x)}{2} + \cos(6) + \cos(x^2)}$$

two kinds of ~~sub~~ reevaluation

$x^2 \to y \qquad \cos \to \sin$